Quantitative Models & Mathematical Framework

Crypto HFT Tool

A comprehensive reference of all statistical, financial, and risk models implemented in the trading system.

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1. Statistical Arbitrage — Mean-Reversion Signals

1.1 Rolling Z-Score

The core signal measures how far the current cross-exchange spread deviates from its rolling mean, expressed in standard-deviation units. A high absolute z-score indicates a potential mean-reversion opportunity.

$$z = \frac{x_t - \bar{x}_w}{\sigma_w} \tag{1}$$

- x_t — Current spread observation (bid₁ − ask₂)
- $\bar{x}_w$ — Rolling mean of spread over window w
- σ_w — Rolling sample standard deviation (ddof = 1) over window w

Multiple window lengths are evaluated simultaneously — w ∈ {15, 60, 180, 360} — to capture signals across time horizons.

1.2 Volatility-Adjusted Z-Score

The raw z-score is scaled by a volatility adjustment factor that compares recent volatility to full-window volatility, preventing false signals during regime changes.

$$z_{adj} = \frac{x_t - \bar{x}_w}{\sigma_w \cdot \alpha_v} \tag{2}$$

$$\alpha_v = \text{clip}\left(\frac{\sigma_{30}}{\sigma_w},\ \ 0.5,\ \ 2.0\right) \tag{3}$$

- σ_{30} — Standard deviation of the most recent 30 observations
- α_v — Volatility ratio, clamped to [0.5, 2.0]

1.3 Volume-Weighted Z-Score

Incorporates trade volume to weight the signal. High-volume deviations are more meaningful than low-volume ones.

$$z_{vw} = z \cdot \frac{V_t}{\bar{V}} \tag{4}$$

- V_t — Current period trade volume
- $\bar{V}$ — Rolling average volume over the lookback window

1.4 Spread Returns & Momentum

Spread returns and momentum are computed to capture directional persistence in the spread series.

$$r_t = \frac{s_t - s_{t-1}}{s_{t-1}} \tag{5}$$

$$M = \sum_{i=t-k}^{t} r_i \tag{6}$$

- s_t — Spread at time t
- r_t — Spread return at time t

M — Momentum score over k recent periods

2. Adaptive Signal Thresholds

Rather than static z-score entry/exit thresholds, the system dynamically adjusts thresholds based on current market conditions across four dimensions.

$$\tau = \tau_0 \cdot A_{vol} \cdot A_{liq} \cdot A_{time} \cdot (1 + A_{mom}) \tag{7}$$

Volatility adjustment:

$$A_{vol} = 1 + \sigma_{ann} \cdot \lambda_v \tag{8}$$

Liquidity (volume) adjustment:

$$A_{liq} = \frac{1}{1 + \ln(1 + V_t / \bar{V})} \tag{9}$$

Momentum impact:

$$A_{mom} = |M| \cdot 0.1 \tag{10}$$

τ_0 — Base threshold from configuration

λ_v — Volatility impact scaling factor

A_{time} — Time-of-day factor (configurable)

The final threshold is clamped: τ ∈ [0.5, 5.0].

3. Volatility & Correlation Estimation

3.1 Annualized Volatility

Return volatility is annualized using the square-root-of-time rule, assuming 252 trading days per year.

$$\sigma_{ann} = \mathrm{std}(r_1, r_2, \ldots, r_n) \cdot \sqrt{252} \tag{11}$$

σ_{ann} — Annualized volatility of price or spread returns

3.2 Cross-Asset Correlation

Pearson correlation between return series of different trading pairs is used for portfolio-level risk management.

$$\rho_{ij} = \frac{\mathrm{Cov}(r_i, r_j)}{\sigma_i \cdot \sigma_j} \tag{12}$$

ρ_{ij} — Pearson correlation between assets i and j

r_i, r_j — Return series of the two assets

4. Risk Management Framework

4.1 Position Sizing Model

Position size is dynamically adjusted through a multiplicative model incorporating four independent risk factors.

$$Q = Q_0 \cdot f_{vol} \cdot f_{sig} \cdot f_{heat} \cdot f_{corr} \tag{13}$$

Volatility factor — reduces size in high-volatility environments:

$$f_{vol} = \frac{1}{1 + \sigma_{ann} \cdot \kappa} \tag{14}$$

Signal strength factor — square-root scaling for diminishing returns:

$$f_{sig} = \sqrt{S} \tag{15}$$

Portfolio heat factor — reduces exposure as aggregate risk grows:

$$f_{heat} = \frac{1}{1 + H} \tag{16}$$

Correlation factor — diversification penalty:

$$f_{corr} = \frac{1}{1 + \bar{\rho}} \tag{17}$$

- Q_0 — Base position size from configuration
- κ — Volatility scaling factor
- S — Signal strength ∈ [0, 1]
- H — Portfolio heat (defined in §4.2)
- $\bar{\rho}$ — Average correlation across active positions

4.2 Portfolio Heat

Aggregate portfolio risk exposure, expressed as a fraction of notional portfolio value.

$$H = \frac{\sum_i |q_i \cdot p_i|}{V_{portfolio}} \tag{18}$$

- q_i — Position size in asset i
- p_i — Current price of asset i
- $V_{portfolio}$ — Base portfolio value

4.3 Stop-Loss Trigger

A position is liquidated when the spread moves adversely beyond a configured percentage threshold.

$$\Delta_{stop} = \frac{|s_t - s_{entry}|}{|s_{entry}|} > \theta_{SL} \tag{19}$$

- s_{entry} — Spread at position entry
- θ_{SL} — Stop-loss threshold (default 2%)

4.4 Drawdown Monitoring

Maximum drawdown tracks the largest peak-to-trough decline in cumulative PnL, used to halt trading if limits are breached.

$$DD_t = \frac{P_t^{peak} - PnL_t^{cum}}{|P_t^{peak}|} \tag{20}$$

P_t^{peak} — Running peak of cumulative PnL up to time t

PnL_t^{cum} — Cumulative realized PnL at time t

4.5 Value-at-Risk (VaR) Limit

Before opening a new position, the system verifies that total exposure does not exceed a 1% daily VaR limit.

$$VaR_{limit} = 0.01 \cdot (V_{portfolio} + V_{new}) \tag{21}$$

V_{new} — Notional value of the proposed new position

5. Order Book Analytics & Execution

5.1 Mid Price & Spread

The mid price and cross-exchange spread form the foundation of all arbitrage calculations.

$$P_{mid} = \frac{P_{bid} + P_{ask}}{2} \tag{22}$$

$$S = P_{bid}^{A} - P_{ask}^{B} \tag{23}$$

P_{bid}^{A} — Best bid on exchange A

P_{ask}^{B} — Best ask on exchange B

5.2 Spread Ratio

Normalised spread used for threshold-based filtering.

$$R_{spread} = \frac{|S|}{P_{avg}} \tag{24}$$

P_{avg} — Average of bid and ask across exchanges

5.3 Liquidity Score

Measures whether sufficient order-book depth exists to fill the desired trade size.

$$L = \min\left(1,\ \frac{\sum_k d_k}{Q_{desired}}\right) \tag{25}$$

d_k — Quantity available at level k in the order book

$Q_{desired}$ — Target fill quantity

5.4 Price Impact (VWAP-Based)

Estimated cost of walking the order book, measured as percentage deviation from the best available price.

$$P_{vwap} = \frac{\sum_k p_k \cdot q_k}{\sum_k q_k} \tag{26}$$

$$I = \frac{|P_{vwap} - P_{best}|}{P_{best}} \tag{27}$$

p_k, q_k — Price and quantity at order-book level k

P_{best} — Best bid or ask (reference price)

I — Fractional price impact

5.5 Slippage

Post-execution slippage comparing the expected fill price to the actual executed price.

$$\epsilon = \frac{|P_{exec} - P_{expected}|}{P_{expected}} \tag{28}$$

P_{exec} — Actual volume-weighted execution price

$P_{expected}$ — Expected price at order submission

5.6 Iceberg Order Sizing

Large orders are split into smaller chunks to minimise market impact. Chunk size is the lesser of 20% of the order or 10% of visible book depth.

$$C = \min(0.2 \cdot Q,\ \ 0.1 \cdot D_{total}) \tag{29}$$

C — Chunk size per child order

D_{total} — Total visible depth in the order book

6. Fee-Adjusted PnL Calculation

6.1 Effective Price After Fees

Execution prices are adjusted for exchange trading fees to reflect true cost basis.

$$P_{eff}^{buy} = P \cdot (1 + \phi) \tag{30}$$

$$P_{eff}^{sell} = P \cdot (1 - \phi) \tag{31}$$

ϕ — Exchange fee rate (maker or taker)

6.2 Arbitrage Net PnL

The net profit from a completed arbitrage round-trip after deducting fees on both legs.

$$PnL = Q \cdot P_{sell} - Q \cdot P_{buy} - F_{buy} - F_{sell} \tag{32}$$

$$F = Q \cdot P \cdot \phi \tag{33}$$

F_{buy}, F_{sell} — Fee cost on the buy and sell legs

6.3 Position Average Price (VWAP)

When adding to an existing position, the average entry price is updated as a volume-weighted average.

$$\bar{P}_{new} = \frac{q_{old} \cdot \bar{P}_{old} + q_{add} \cdot P_{add}}{q_{old} + q_{add}} \tag{34}$$

7. Backtest Performance Metrics

7.1 Win Rate

$$W = \frac{N_{win}}{N_{total}} \times 100\% \tag{35}$$

N_{win} — Number of trades with positive PnL

N_{total} — Total number of closed trades

7.2 Annualized Sharpe Ratio

Risk-adjusted return metric assuming zero risk-free rate, annualized with √252.

$$SR = \frac{\bar{r}}{\sigma_r} \cdot \sqrt{252} \tag{36}$$

$\bar{r}$ — Mean per-trade PnL

σ_r — Standard deviation of per-trade PnL

7.3 Maximum Drawdown

The largest peak-to-trough percentage decline in the cumulative equity curve during the backtest period.

$$MDD = \min_t \left(\frac{C_t - \max_{\tau \le t} C_\tau}{\max_{\tau \le t} C_\tau} \right) \tag{37}$$

C_t — Cumulative PnL at time t

8. Price Simulation (Sample Data Generation)

8.1 Geometric Random Walk

Sample price data is generated via a geometric random walk model with normally distributed log-returns.

$$P_t = P_0 \prod_{i=1}^{t} (1 + \epsilon_i), \quad \epsilon_i \sim \mathcal{N}(0,\ 0.001) \tag{38}$$

8.2 Bid-Ask Construction

Simulated bid and ask prices are constructed symmetrically around a mid price with a configurable half-spread.

$$P_{bid} = P_{mid} \cdot \left(1 - \frac{\delta}{2}\right) \tag{39}$$

$$P_{ask} = P_{mid} \cdot \left(1 + \frac{\delta}{2}\right) \tag{40}$$

δ — Spread percentage parameter

This document was auto-generated from static analysis of the crypto_hft_tool source code. Equations (1)–(40) represent the complete quantitative framework of the system.